



KTU NOTES

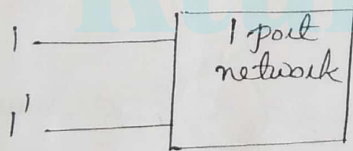
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Two-port network

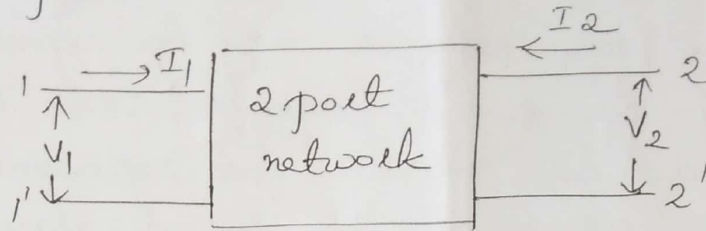
Generally any network may be represented schematically by a rectangular box. A network may be used for representing either source or load or for a variety of purposes. A pair of terminals at which a signal may enter or leave a network is called a port. A port is defined as any pair of terminals into which energy is supplied, or from which energy is withdrawn or where the network variables may be measured. One such network having only one pair of terminals ($1-1'$) is shown below. It is called one port network.



A two-port network is simply a network inside a black box, and the network has only two pairs of accessible terminals; usually one pair represents the input and the other represents the output. Such a building block is very common in electronic systems, communication systems, transmission and distribution systems.

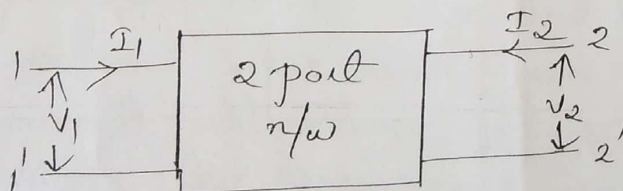
Figure below shows a two-port n/w two terminal pair network, in

which the four terminals have been paired into ports 1-1' and 2-2'.



The terminals 1-1' together constitute a port. Similarly, the terminals 2-2' constitutes another port. A voltage and current is assigned to each of the two ports. The voltage and current at the input terminals are V_1 and I_1 , whereas V_2 and I_2 are specified at the output port. It is also assumed that the currents I_1 and I_2 are entering into the network at the upper terminals 1 and 2 respectively.

Open circuit impedance (Z) parameters



$$[V] = [Z] [I]$$

$$\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix}$$

$$V_1 = Z_{11} I_1 + Z_{12} I_2$$

$$V_2 = Z_{21} I_1 + Z_{22} I_2$$

(2)

Z_{11} , Z_{12} , Z_{21} and Z_{22} are the network functions and are called impedance (Z) parameters.

The individual Z parameters for a given n/w can be defined by setting each of the port currents equal to zero.

Suppose port 2-2' is open circuited, $I_2 = 0$

$$Z_{11} = \frac{V_1}{I_1} \bigg|_{I_2=0}$$

It is called input driving point impedance or open circuit input impedance.

$$Z_{21} = \frac{V_2}{I_1} \bigg|_{I_2=0}$$

Z_{21} is called forward transfer impedance.

If port 1-1' is open circuited, $I_1 = 0$

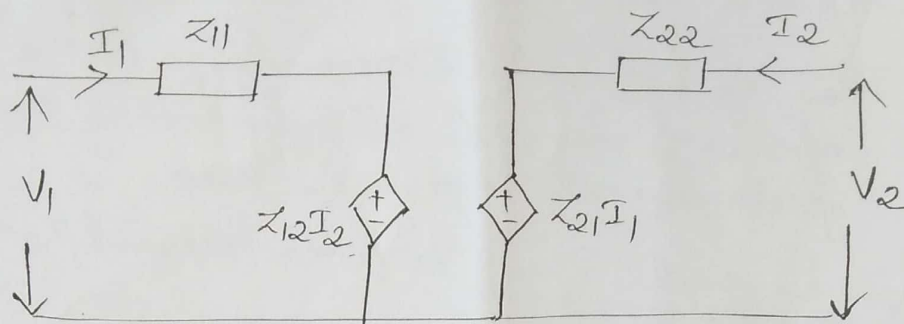
$$Z_{12} = \frac{V_1}{I_2} \bigg|_{I_1=0}$$

Z_{12} is called reverse transfer impedance.

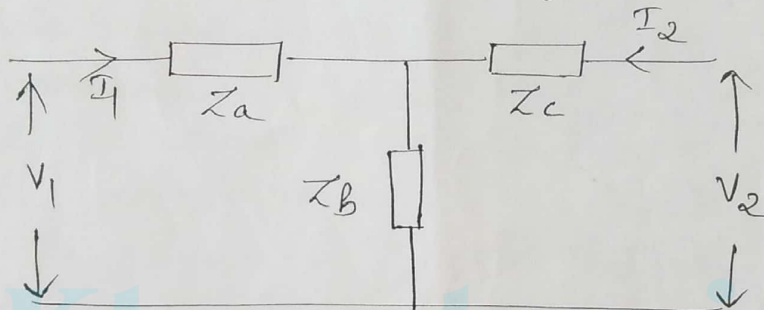
$$Z_{22} = \frac{V_2}{I_2} \bigg|_{I_1=0}$$

Z_{22} is called the output driving point impedance or open circuit output impedance

Equivalent circuit



1. Find the Z parameters of the circuit shown.



sol: Apply KVL in mesh ①,

$$V_1 = (Z_a + Z_b)I_1 + Z_b I_2 \quad \text{--- ①}$$

Apply KVL in mesh ②,

$$\begin{aligned} V_2 &= (Z_b + Z_c)I_2 + Z_b I_1 \\ &= Z_b I_1 + (Z_b + Z_c)I_2 \quad \text{--- ②} \end{aligned}$$

Compare the KVL eqns with

$$\begin{aligned} V_1 &= Z_{11}I_1 + Z_{12}I_2 \\ V_2 &= Z_{21}I_1 + Z_{22}I_2 \end{aligned}$$

$$\therefore Z_{11} = Z_a + Z_b \quad \Omega$$

$$Z_{12} = Z_b \quad \Omega$$

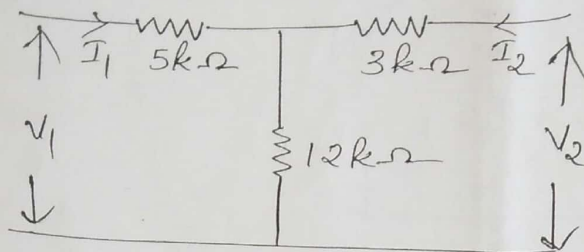
$$Z_{21} = Z_b \quad \Omega$$

$$Z_{22} = Z_b + Z_c \quad \Omega$$



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2. Find the z parameters.

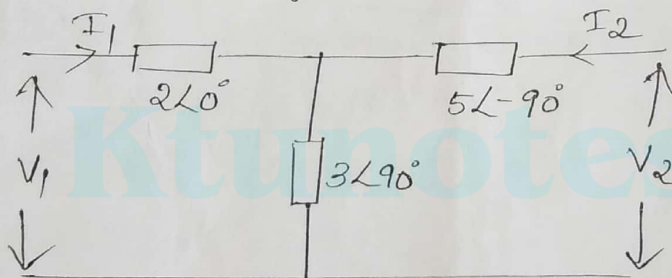


Ans:- $Z_{11} = Z_a + Z_b = 17k\Omega$

$Z_{12} = Z_{21} = 12k\Omega$

$Z_{22} = Z_b + Z_c = \underline{15k\Omega}$

3. Find the z parameters.



Sol:- Writing KVL eqns,

$$V_1 = 2\angle 0^\circ I_1 + 3\angle 90^\circ (I_1 + I_2)$$

$$= (2\angle 0^\circ + 3\angle 90^\circ) I_1 + 3\angle 90^\circ I_2 \quad \text{--- (1)}$$

$$V_2 = 5\angle -90^\circ I_2 + 3\angle 90^\circ (I_1 + I_2)$$

$$= 3\angle 90^\circ I_1 + I_2 (5\angle -90^\circ + 3\angle 90^\circ) \quad \text{--- (2)}$$

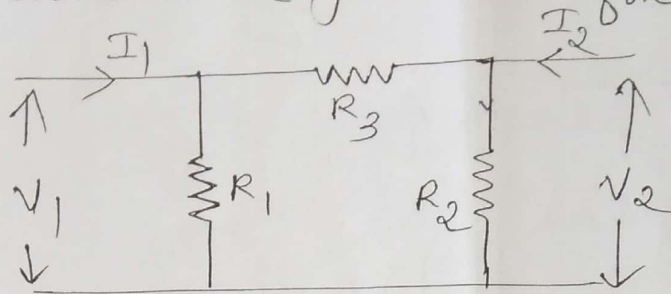
$\therefore Z_{11} = 2 + 3j = 3.6\angle 56.3^\circ \Omega$

$Z_{12} = 3\angle 90^\circ \Omega$

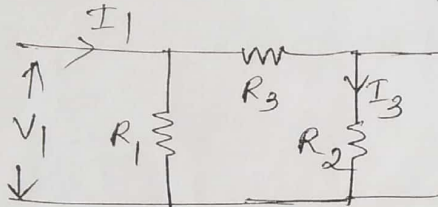
$Z_{21} = 3\angle 90^\circ \Omega$

$Z_{22} = 2\angle -90^\circ \Omega$

4) Determine z parameter for the π network.



Sol: open circuit output port, $I_2 = 0$



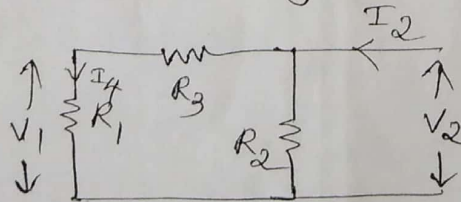
$$Z_{11} = \frac{V_1}{I_1} = \frac{(R_2 + R_3) \parallel R_1 \cdot I_1}{I_1} = \frac{R_1 (R_2 + R_3)}{R_1 + R_2 + R_3} \Omega$$

$$Z_{21} = \frac{V_2}{I_1} = \frac{I_3 \cdot R_2}{I_1}$$

$$\text{Here, } I_3 = \frac{I_1 \cdot R_1}{R_1 + R_2 + R_3} \quad \left[\text{By current division} \right]$$

$$\therefore Z_{21} = \frac{R_1 R_2}{R_1 + R_2 + R_3} \Omega$$

open circuit input port, $I_1 = 0$



$$Z_{22} = \frac{V_2}{I_2} = (R_1 + R_3) \parallel R_2 \cdot \frac{I_2}{I_2}$$

$$= \frac{(R_1 + R_3) \cdot R_2}{R_1 + R_2 + R_3} \Omega$$

$$Z_{12} = \frac{V_1}{I_2} = \frac{R_1 I_4}{I_2}$$

$$\text{Here, } I_4 = \frac{I_2 \cdot R_2}{R_1 + R_2 + R_3}$$

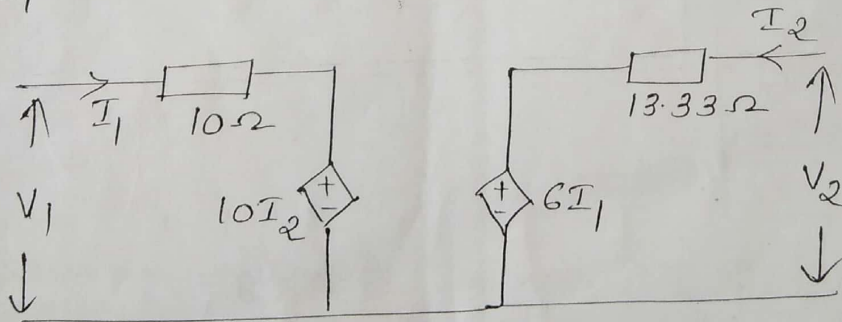
$$\therefore Z_{12} = \frac{R_1 R_2}{R_1 + R_2 + R_3} \Omega$$

5. Draw the Z parameter equivalent circuit.
 $Z_{11} = 10 \Omega$, $Z_{12} = 10 \Omega$, $Z_{21} = 6 \Omega$, $Z_{22} = 13.33 \Omega$

sol:- $V_1 = 10I_1 + 10I_2$

$$V_2 = 6I_1 + 13.33I_2$$

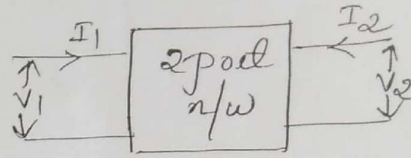
$10I_2$ and $6I_1$ are the controlled sources and
 $10I_1$ and $13.33I_2$ are the voltage drops.



Short circuit admittance (Y) parameters

$$[I] = [Y][V]$$

$$\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix}$$



$$I_1 = Y_{11}V_1 + Y_{12}V_2$$

$$I_2 = Y_{21}V_1 + Y_{22}V_2$$

Assume output port is short circuited, $V_2 = 0$

$$\boxed{\begin{aligned} Y_{11} &= \frac{I_1}{V_1} \\ Y_{21} &= I_2/V_1 \end{aligned}}$$

Y_{11} is called input driving point admittance

Y_{21} is called forward transfer admittance.

Its unit is mho (Ω^{-1}).

Assume input port is short circuited, $V_1 = 0$

$$\boxed{\begin{aligned} Y_{12} &= \frac{I_1}{V_2} \\ Y_{22} &= \frac{I_2}{V_2} \end{aligned}}$$

Y_{12} is called reverse transfer admittance.

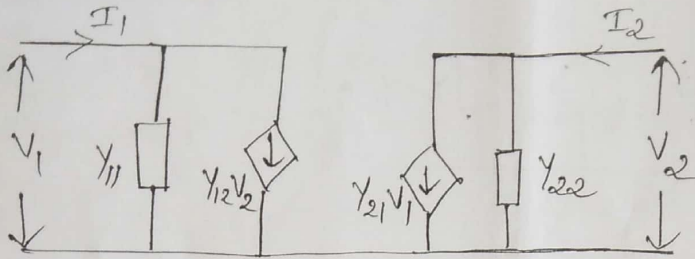
Y_{22} is called output driving point admittance.

Unit is Ω^{-1} .

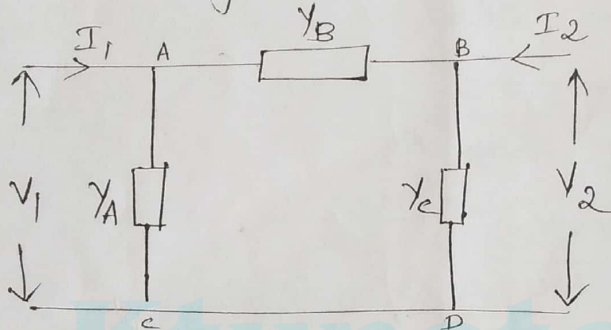


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Equivalent circuit representation



Find the Y parameters and draw the eq^t. ckt.



Apply KCL at node A,

$$\begin{aligned} I_1 &= V_1 Y_A + (V_1 - V_2) Y_B \\ &= (Y_A + Y_B) V_1 - Y_B V_2 \quad \text{--- (1)} \end{aligned}$$

Apply KCL at node B,

$$\begin{aligned} I_2 &= V_2 Y_c + (V_2 - V_1) Y_B \\ &= -Y_B V_1 + (Y_c + Y_B) V_2 \quad \text{--- (2)} \end{aligned}$$

Comparing with standard eqns, we get.

$$Y_{11} = Y_A + Y_B \quad \checkmark$$

$$Y_{12} = -Y_B \quad \checkmark$$

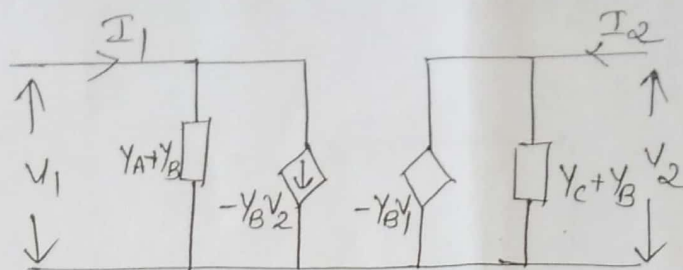
$$Y_{21} = -Y_B \quad \checkmark$$

$$Y_{22} = Y_c + Y_B \quad \checkmark$$

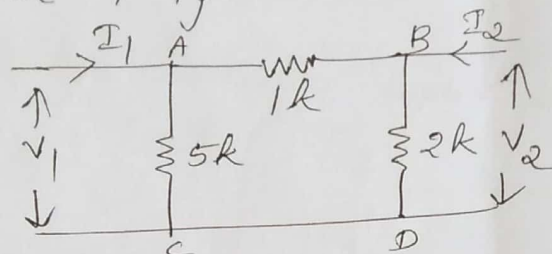


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Equivalent circuit is,



2. Find Y parameters



so! Apply KCL at node A,

$$I_1 = \frac{V_1}{5k} + \frac{(V_1 - V_2)}{1k}$$

$$= V_1 \left(\frac{1}{5k} + \frac{1}{1k} \right) - \frac{V_2}{1k} \quad \text{--- (1)}$$

Apply KCL at node B,

$$I_2 = \frac{V_2}{2k} + \frac{(V_2 - V_1)}{1k}$$

$$= -\frac{V_1}{1k} + \left(\frac{1}{2k} + \frac{1}{1k} \right) V_2 \quad \text{--- (2)}$$

Comparing we get,

$$Y_{11} = 1.2 \times 10^{-3} \text{ mho}$$

$$Y_{12} = -10^{-3} \text{ v}$$

$$Y_{21} = -10^{-3} \text{ v}$$

$$Y_{22} = \underline{\underline{1.5 \times 10^{-3} \text{ v}}}$$

3. Following short ckt currents and voltages are obtained experimentally for a 2 port n/w

a) With o/p short circuited,

$$I_1 = 5 \text{ mA}, I_2 = -0.3 \text{ mA}, V_1 = 25 \text{ V}.$$

b) With i/p short circuited,

$$I_1 = -5 \text{ mA}, I_2 = 10 \text{ mA}, V_2 = 30 \text{ V}$$

Determine Y parameters.

1:- With o/p short cktd,

$$Y_{11} = \frac{I_1}{V_1} = \frac{5 \text{ mA}}{25 \text{ V}} = \underline{\underline{0.0002 \text{ S}}}$$

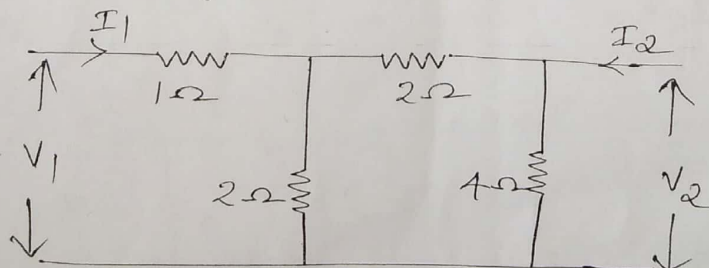
$$Y_{21} = \frac{I_2}{V_1} = \frac{-0.3 \times 10^{-3}}{25} = \underline{\underline{-0.000012 \text{ S}}}$$

With i/p short cktd,

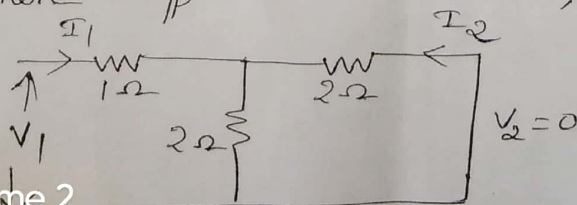
$$Y_{12} = \frac{I_1}{V_2} = \frac{-5 \times 10^{-3}}{30} = \underline{\underline{-0.00017 \text{ S}}}$$

$$Y_{22} = \frac{I_2}{V_2} = \frac{10 \times 10^{-3}}{30} = \underline{\underline{0.00033 \text{ S}}}$$

4. Find Y parameters for the following circuit.



sol:- When o/p is short cktd, $V_2 = 0$



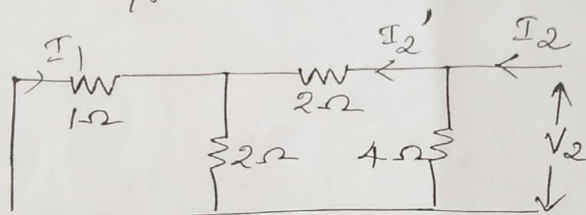
$$Y_{11} = \frac{I_1}{V_1} = \frac{I_1}{R_{eq} \cdot I_1} = \frac{I_1}{2I_1} = \underline{\underline{\frac{1}{2} \Omega^{-1}}}$$

$$Y_{21} = \frac{I_2}{V_1}$$

$$I_2 = -I_1 \times \frac{2}{4} \quad [\because \text{current division}]$$

$$\therefore Y_{21} = \frac{-I_1/2}{2I_1} = \underline{\underline{-1/4 \Omega^{-1}}}$$

When i/p is short ckt'd, $V_1 = 0$



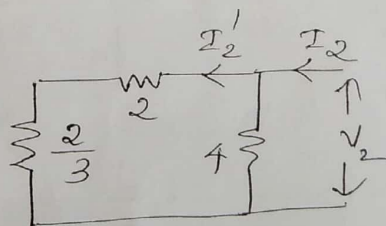
$$\begin{aligned} Y_{22} &= \frac{I_2}{V_2} = \frac{I_2}{R_{eq} \cdot I_2} \\ &= \frac{I_2}{\frac{8}{5} \cdot I_2} \\ &= \underline{\underline{\frac{5}{8} \Omega^{-1}}} \end{aligned}$$

$$\begin{aligned} R_{eq} &= ((1 \parallel 2) + 2) \parallel 4 \\ &= \frac{8}{5} \Omega \end{aligned}$$

$$Y_{12} = \frac{I_1}{V_2}$$

$$I_1 = -\frac{I_2' \cdot 2}{3}$$

$$\begin{aligned} I_2' &= \frac{I_2 \cdot 4}{4 + 2 + \frac{2}{3}} = \frac{I_2 \cdot 4}{\frac{20}{3}} \\ &= \frac{12}{20} I_2 = \frac{3}{5} I_2 \end{aligned}$$



$$I_1 = -\frac{\frac{3}{5} I_2 \cdot 2}{3} = -\frac{6}{15} I_2 = -\frac{2}{5} I_2$$

$$Y_{12} = \frac{I_1}{V_2} = -\frac{\frac{2}{5} I_2}{\frac{8}{5} I_2} = -\frac{2}{8} = \underline{\underline{-1/4 \Omega^{-1}}}$$

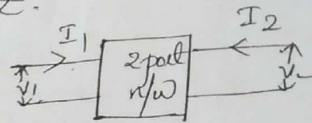
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Hybrid (h) Parameters

h parameter representation is widely used in modelling of electronic components and circuits, particularly transistors. As both short circuit and open circuit terminal conditions are utilized, this parameter is called hybrid parameters.

In this form, voltage of the i/p port and current of the o/p port are expressed in terms of the current of the i/p port and voltage of the o/p port.

$$\begin{bmatrix} V_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ V_2 \end{bmatrix}$$



$$V_1 = h_{11} I_1 + h_{12} V_2$$

$$I_2 = h_{21} I_1 + h_{22} V_2$$

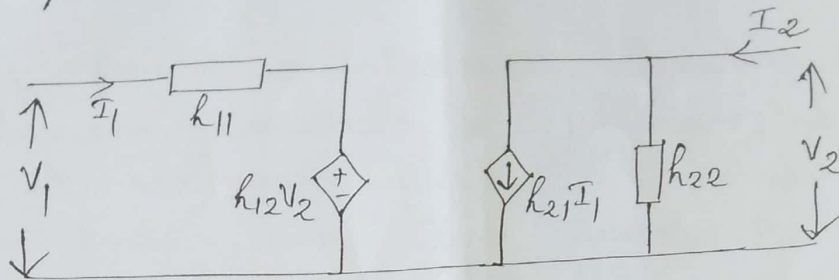
Assuming short ckt at the o/p, $V_2 = 0$

$$\begin{bmatrix} h_{11} = \frac{V_1}{I_1} \rightarrow \text{Input Impedance } (\Omega) \\ h_{21} = \frac{I_2}{I_1} \rightarrow \text{Forward current gain (unitless)} \end{bmatrix}$$

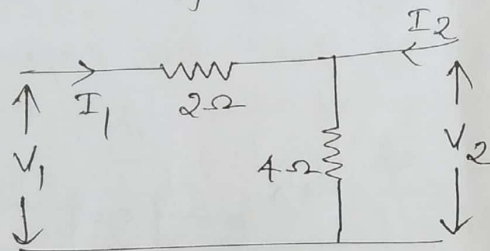
Assuming open circuit at the i/p, $I_1 = 0$

$$\begin{bmatrix} h_{12} = \frac{V_1}{V_2} \rightarrow \text{Reverse voltage gain (unitless)} \\ h_{22} = \frac{I_2}{V_2} \rightarrow \text{output admittance } (\Omega^{-1}) \end{bmatrix}$$

Equivalent circuit



1. Find R parameters.



sol: Short ckt o/p port, $V_2 = 0$

$$h_{11} = \frac{V_1}{I_1}$$

$$= \frac{2I_1}{I_1} = \underline{\underline{2\Omega}}$$

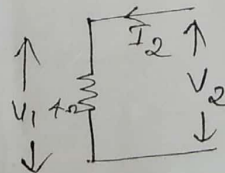
$$h_{21} = \frac{I_2}{I_1} = \frac{-I_1}{I_1} = \underline{\underline{-1}}$$

open ckt i/p port, $I_1 = 0$

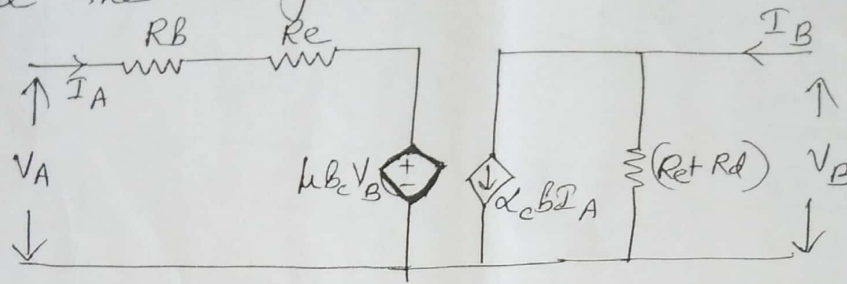
$$h_{12} = \frac{V_1}{V_2} = \frac{4I_2}{4I_2} = \underline{\underline{1}}$$

$$h_{22} = \frac{I_2}{V_2} = \frac{I_2}{4I_2}$$

$$= \underline{\underline{\frac{1}{4} \Omega}}$$



2. CE transistor equivalent ckt is shown below.
Find the h parameters.



sol: KVL at i/p port,

$$V_A = (R_B + R_E) I_A + h_{be} V_B \quad \text{--- (1)}$$

KCL at o/p port,

$$I_B = \frac{V_B}{R_E + R_D} + \alpha_{cb} I_A \quad \text{--- (2)}$$

$$\begin{bmatrix} V_A \\ I_B \end{bmatrix} = \begin{bmatrix} R_B + R_E & h_{be} \\ \alpha_{cb} & \frac{1}{R_E + R_D} \end{bmatrix} \begin{bmatrix} I_A \\ V_B \end{bmatrix}$$

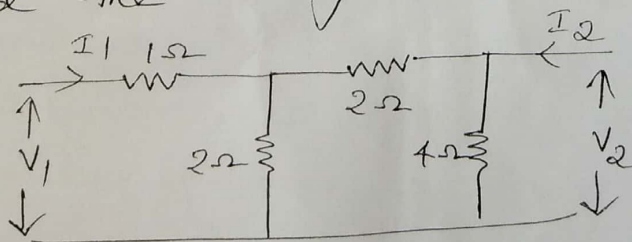
$$\therefore h_{11} = R_B + R_E$$

$$h_{12} = h_{be}$$

$$h_{21} = \alpha_{cb}$$

$$h_{22} = \frac{1}{R_E + R_D}$$

3. Find the h parameters.



R

Shot on realme 2

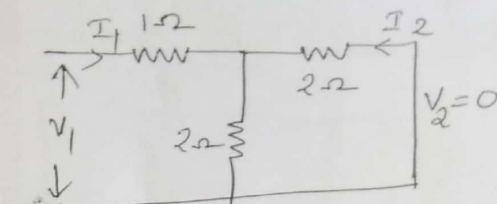
Sol:- short ckt o/p port, $V_2 = 0$.

$$h_{11} = \frac{V_1}{I_1} = \frac{2I_1}{I_1}$$

$$= \underline{\underline{2\Omega}}$$

$$h_{21} = \frac{I_2}{I_1} = \frac{-I_1 \cdot 2}{I_1}$$

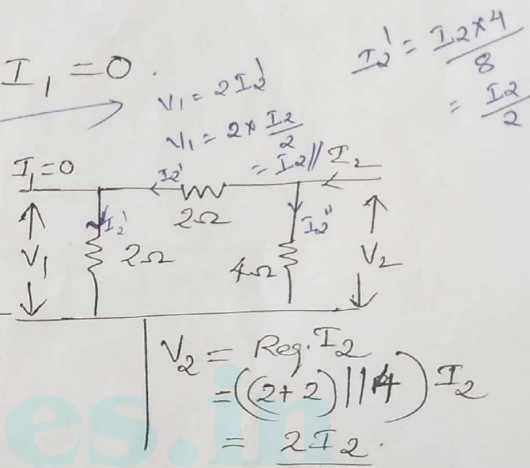
$$= \underline{\underline{-1/2}}$$



open ckt i/p port, $I_1 = 0$.

$$h_{12} = \frac{V_1}{V_2} = \frac{8I_2}{4I_2} = \underline{\underline{1/2}}$$

$$h_{22} = \frac{I_2}{V_2} = \frac{I_2}{2I_2} = \underline{\underline{1/2}}$$

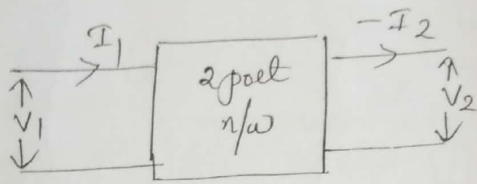


ABCD parameters (Circuit parameters, Transmission parameters)

ABCD parameters are widely used in analysis of power transmission engineering where they are termed as circuit parameters. ABCD parameters are also known as transmission parameters (T parameter). It is conventional to designate the i/p port as sending end and the o/p port as receiving end while representing ABCD parameters.

R

Shot on realme 2 (Note: o/p current direction is taken reverse)



$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} V_2 \\ -I_2 \end{bmatrix}$$

$$V_1 = AV_2 - BI_2$$

$$I_1 = CV_2 - DI_2$$

Assume receiving end to be open-cktd, $I_2 = 0$

$$A = \frac{V_1}{V_2} \rightarrow \text{reverse voltage gain (unitless)}$$

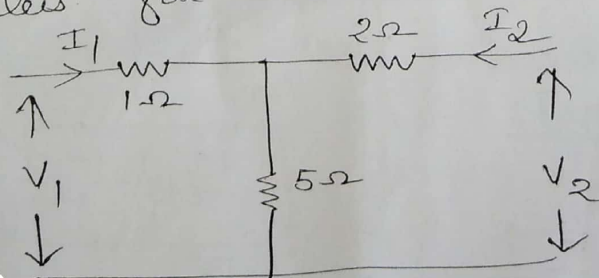
$$C = \frac{I_1}{V_2} \rightarrow \text{Transfer admittance (S)}$$

Assume receiving end to be short-cktd, $V_2 = 0$

$$B = \frac{V_1}{-I_2} \rightarrow \text{Transfer impedance (}\Omega\text{)}$$

$$D = \frac{I_1}{-I_2} \rightarrow \text{Reverse current gain (unitless)}$$

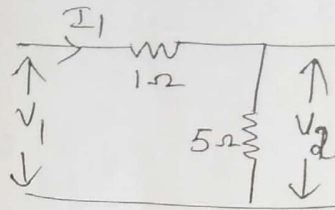
- Find the transmission or general ckt parameters for the ckt shown below.



Sol: open ckt receiving end, $I_2 = 0$.

$$A = \frac{V_1}{V_2} = \frac{6I_1}{5I_1} = \underline{\underline{\frac{6}{5}}}$$

$$C = \frac{I_1}{V_2} = \frac{I_1}{5I_1} = \underline{\underline{\frac{1}{5} \text{ V}}}$$



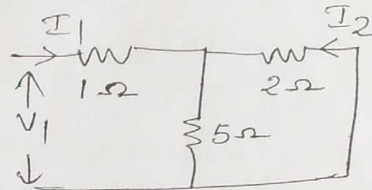
Short ckt receiving end, $V_2 = 0$

$$B = \frac{V_1}{-I_2} = \frac{R_{eq} \cdot I_1}{-(-I_1 \cdot \frac{5}{7})}$$

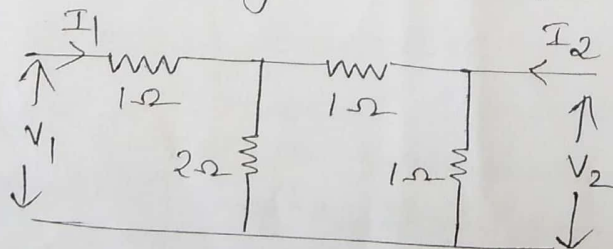
$$= \frac{((2//5) + 1) I_1}{I_1 \cdot \frac{5}{7}}$$

$$= \frac{\frac{17}{7} I_1}{\frac{5}{7} I_1} = \underline{\underline{\frac{17}{5} \Omega}}$$

$$D = \frac{I_1}{-I_2} = \frac{I_1}{I_1 \cdot \frac{5}{7}} = \underline{\underline{\frac{7}{5}}}$$



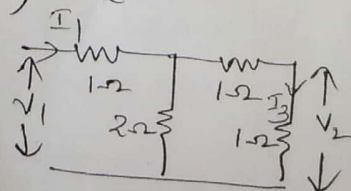
2. Find ABCD parameters for the n/w shown below.



Sol: open ckt receiving end, $I_2 = 0$.

$$A = \frac{V_1}{V_2} = \frac{R_{eq} \cdot I_1}{1 \cdot I_3}$$

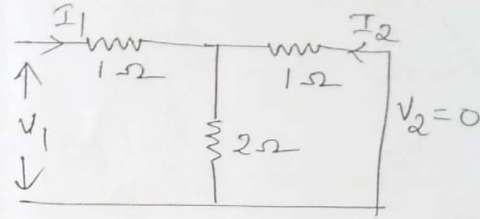
$$= \frac{2 I_1}{I_1 \cdot \frac{2}{4}} = \underline{\underline{4}}$$



$$C = \frac{I_1}{V_2} = \frac{I_1}{I_1/2} = \underline{\underline{2\Omega}}$$

short ckt receiving end, $V_2=0$.

$$\begin{aligned} B &= \frac{V_1}{-I_2} \\ &= \frac{\frac{5}{3} I_1}{\frac{I_1 \cdot 2}{3}} = \underline{\underline{\frac{5}{2} \Omega}} \end{aligned}$$



$$\begin{aligned} D &= \frac{I_1}{-I_2} \\ &= \frac{I_1}{\frac{I_1 \cdot 2}{3}} = \underline{\underline{\frac{3}{2}}} \end{aligned}$$

Inter-relationships b/w parameter sets

1. Z parameter in terms of Y parameter

$$\begin{aligned} Z &= Y^{-1} \\ \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} &= \begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix}^{-1} = \frac{\text{Adj}[Y]}{|Y|} = \frac{[\text{Co-factor } Y]^T}{|Y|} \\ &= \frac{\begin{bmatrix} Y_{22} & -Y_{12} \\ -Y_{21} & Y_{11} \end{bmatrix}^T}{\begin{vmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{vmatrix}}} = \begin{bmatrix} \frac{Y_{22}}{\Delta Y} & \frac{-Y_{12}}{\Delta Y} \\ \frac{-Y_{21}}{\Delta Y} & \frac{Y_{11}}{\Delta Y} \end{bmatrix} \end{aligned}$$

$$Z_{11} = Y_{22}/\Delta Y$$

$$Z_{12} = -Y_{12}/\Delta Y$$

$$Z_{21} = -Y_{21}/\Delta Y$$

$$Z_{22} = Y_{11}/\Delta Y$$

2. Z parameter in terms of h parameter

$$V_1 = h_{11} I_1 + h_{12} V_2 \quad \text{--- (1)}$$

$$I_2 = h_{21} I_1 + h_{22} V_2 \quad \text{--- (2)}$$

$$\left| \begin{array}{l} V_1 = Z_{11} I_1 + Z_{12} I_2 \\ V_2 = Z_{21} I_1 + Z_{22} I_2 \end{array} \right.$$

From (2),

$$V_2 = -\frac{h_{21}}{h_{22}} I_1 + \frac{1}{h_{22}} I_2$$

Sub. this in eqn (1) $\Rightarrow$

$$V_1 = h_{11} I_1 + h_{12} \left[-\frac{h_{21}}{h_{22}} I_1 + \frac{1}{h_{22}} I_2 \right]$$

$$= \left[h_{11} - \frac{h_{21} h_{12}}{h_{22}} \right] I_1 + \frac{h_{12}}{h_{22}} I_2$$

$$\therefore Z_{11} = \frac{h_{11} h_{22} - h_{12} h_{21}}{h_{22}} = \frac{\Delta h}{h_{22}}$$

$$Z_{12} = \frac{h_{12}}{h_{22}}$$

$$Z_{21} = -\frac{h_{21}}{h_{22}}$$

$$Z_{22} = \frac{1}{h_{22}}$$

3. Z parameter in terms of ABCD parameters

$$V_1 = A V_2 - B I_2 \quad \text{--- (1)}$$

$$I_1 = C V_2 - D I_2 \quad \text{--- (2)}$$

$$\text{From (2)} \Rightarrow V_2 = \frac{1}{C} I_1 + \frac{D}{C} I_2$$

Sub. in eqn (1) $\Rightarrow$

$$V_1 = A \left(\frac{1}{C} I_1 + \frac{D}{C} I_2 \right) - B I_2$$

$$= \frac{A}{C} I_1 + \left(\frac{AD - BC}{C} \right) I_2$$

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$$\therefore Z_{11} = \frac{A}{C}$$

$$Z_{12} = \frac{AD-BC}{C}$$

$$Z_{21} = \frac{1}{C}$$

$$Z_{22} = \frac{D}{C}$$

4. Y parameters in terms of Z parameters

$$[Y] = [Z]^{-1}$$

$$\begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix} = \begin{bmatrix} Z_{22}/\Delta Z & -Z_{12}/\Delta Z \\ -Z_{21}/\Delta Z & Z_{11}/\Delta Z \end{bmatrix}$$

5. Y parameters in terms of h parameters

$$V_1 = h_{11} I_1 + h_{12} V_2 \quad \text{--- (1)}$$

$$I_2 = h_{21} I_1 + h_{22} V_2 \quad \text{--- (2)}$$

$$\begin{cases} I_1 = Y_{11} V_1 + Y_{12} V_2 \\ I_2 = Y_{21} V_1 + Y_{22} V_2 \end{cases}$$

From (1) $\Rightarrow$

$$I_1 = \frac{1}{h_{11}} V_1 - \frac{h_{12}}{h_{11}} V_2 \quad \text{--- (3)}$$

sub (3) in (2) $\Rightarrow$

$$\begin{aligned} I_2 &= \frac{h_{21}}{h_{11}} V_1 - \frac{h_{12} h_{21}}{h_{11}} V_2 + h_{22} V_2 \\ &= \frac{h_{21}}{h_{11}} V_1 + \frac{\Delta h}{h_{11}} V_2 \quad \text{--- (4)} \end{aligned}$$

From (3) & (4) $\Rightarrow$

$$Y_{11} = \frac{1}{h_{11}}, \quad Y_{12} = -\frac{h_{12}}{h_{11}}$$

$$Y_{21} = \frac{h_{21}}{h_{11}}, \quad Y_{22} = \frac{\Delta h}{h_{11}}$$

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6. Y parameters in terms of ABCD

$$V_1 = AV_2 - BI_2 \quad \text{--- (1)}$$

$$I_1 = CV_2 - DI_2 \quad \text{--- (2)}$$

From (1) $\Rightarrow$

$$I_2 = -\frac{1}{B}V_1 + \frac{A}{B}V_2 \quad \text{--- (3)}$$

Sub (3) in (2) $\Rightarrow$

$$\begin{aligned} I_1 &= CV_2 + \frac{D}{B}V_1 - \frac{AD}{B}V_2 \\ &= \frac{D}{B}V_1 + \frac{BC - AD}{B}V_2 \quad \text{--- (4)} \end{aligned}$$

$$\begin{aligned} Y_{11} &= \frac{D}{B}, & Y_{12} &= \frac{BC - AD}{B} \\ Y_{21} &= -\frac{1}{B}, & Y_{22} &= \frac{A}{B} \end{aligned}$$

7. h parameter in terms of Z

$$V_1 = Z_{11}I_1 + Z_{12}I_2 \quad \text{--- (1)}$$

$$V_2 = Z_{21}I_1 + Z_{22}I_2 \quad \text{--- (2)}$$

$$\begin{cases} V_1 = h_{11}I_1 + h_{12}V_2 \\ I_2 = h_{21}I_1 + h_{22}V_2 \end{cases}$$

From (2) $\Rightarrow$

$$I_2 = -\frac{Z_{21}}{Z_{22}}I_1 + \frac{1}{Z_{22}}V_2 \quad \text{--- (3)}$$

Sub (3) in (1) $\Rightarrow$

$$\begin{aligned} V_1 &= Z_{11}I_1 + \frac{-Z_{12}Z_{21}}{Z_{22}}I_1 + \frac{Z_{12}}{Z_{22}}V_2 \\ &= \left(\frac{Z_{11}Z_{22} - Z_{12}Z_{21}}{Z_{22}} \right) I_1 + \frac{Z_{12}}{Z_{22}}V_2 \quad \text{--- (4)} \end{aligned}$$

From (3) & (4) $\Rightarrow$

$$h_{11} = \frac{\Delta Z}{Z_{22}}, \quad h_{12} = \frac{Z_{12}}{Z_{22}}$$

$$h_{21} = \frac{-Z_{21}}{Z_{22}}, \quad h_{22} = \frac{1}{Z_{22}}$$

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8. h parameters in terms of γ

$$I_1 = \gamma_{11} V_1 + \gamma_{12} V_2 \quad \text{--- (1)}$$

$$I_2 = \gamma_{21} V_1 + \gamma_{22} V_2 \quad \text{--- (2)}$$

From (1) $\Rightarrow$

$$V_1 = \frac{1}{\gamma_{11}} I_1 - \frac{\gamma_{12}}{\gamma_{11}} V_2 \quad \text{--- (3)}$$

Sub (3) in (2) $\Rightarrow$

$$\begin{aligned} I_2 &= \frac{\gamma_{21}}{\gamma_{11}} I_1 - \frac{\gamma_{12}\gamma_{21}}{\gamma_{11}} V_2 + \gamma_{22} V_2 \\ &= \frac{\gamma_{21}}{\gamma_{11}} I_1 + \left(\frac{\gamma_{11}\gamma_{22} - \gamma_{12}\gamma_{21}}{\gamma_{11}} \right) V_2 \quad \text{--- (4)} \end{aligned}$$

From (3) & (4) $\Rightarrow$

$$h_{11} = \frac{1}{\gamma_{11}}, \quad h_{12} = \frac{-\gamma_{12}}{\gamma_{11}}$$

$$h_{21} = \frac{\gamma_{21}}{\gamma_{11}}, \quad h_{22} = \frac{\Delta \gamma}{\gamma_{11}}$$

9. h parameters in terms of ABCD

$$V_1 = AV_2 - BI_2 \quad \text{--- (1)}$$

$$I_1 = CV_2 - DI_2 \quad \text{--- (2)}$$

From (2) $\Rightarrow$

$$I_2 = -\frac{1}{D} I_1 + \frac{C}{D} V_2 \quad \text{--- (3)}$$

Sub (3) in (1) $\Rightarrow$

$$\begin{aligned} V_1 &= AV_2 + \frac{B}{D} I_1 - \frac{BC}{D} V_2 \\ &= \frac{B}{D} I_1 + \frac{AD - BC}{D} V_2 \quad \text{--- (4)} \end{aligned}$$

From (3) & (4) $\Rightarrow$

$$h_{11} = \frac{B}{D}, \quad h_{12} = \frac{AD - BC}{D}$$

$$h_{21} = -\frac{1}{D}, \quad h_{22} = \frac{C}{D}$$



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10. ABCD in terms of Z

$$V_1 = Z_{11} I_1 + Z_{12} I_2 \quad \text{--- (1)}$$

$$V_2 = Z_{21} I_1 + Z_{22} I_2 \quad \text{--- (2)}$$

$$\begin{cases} V_1 = A V_2 - B I_2 \\ I_1 = C V_2 - D I_2 \end{cases}$$

From (2) $\Rightarrow$

$$I_1 = \frac{1}{Z_{21}} V_2 - \frac{Z_{22}}{Z_{21}} I_2 \quad \text{--- (3)}$$

sub (3) in (1) $\Rightarrow$

$$\begin{aligned} V_1 &= \frac{Z_{11}}{Z_{21}} V_2 - \frac{Z_{22} Z_{11}}{Z_{21}} I_2 + Z_{12} I_2 \\ &= \frac{Z_{11}}{Z_{21}} V_2 - \left(\frac{Z_{22} Z_{11} - Z_{12} Z_{21}}{Z_{21}} \right) I_2 \quad \text{--- (4)} \end{aligned}$$

From (3) & (4) $\Rightarrow$

$$A = \frac{Z_{11}}{Z_{21}}$$

$$B = \frac{Z_{22} Z_{11} - Z_{12} Z_{21}}{Z_{21}} = \frac{\Delta Z}{Z_{21}}$$

$$C = \frac{1}{Z_{21}}$$

$$D = \frac{Z_{22}}{Z_{21}}$$

11. ABCD in terms of Y

$$I_1 = Y_{11} V_1 + Y_{12} V_2 \quad \text{--- (1)}$$

$$I_2 = Y_{21} V_1 + Y_{22} V_2 \quad \text{--- (2)}$$

From (2) $\Rightarrow$

$$V_1 = \frac{1}{Y_{21}} I_2 - \frac{Y_{22}}{Y_{21}} V_2 \quad \text{--- (3)}$$

sub (3) in (1) $\Rightarrow$

$$\begin{aligned} I_1 &= \frac{Y_{11}}{Y_{21}} I_2 - \frac{Y_{22} Y_{11}}{Y_{21}} V_2 + Y_{12} V_2 \\ &= \left(\frac{-Y_{22} Y_{11} + Y_{12} Y_{21}}{Y_{21}} \right) V_2 + \frac{Y_{11}}{Y_{21}} I_2 \quad \text{--- (4)} \end{aligned}$$

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From (3) + (4) $\Rightarrow$

$$h_{11} = \frac{-Y_{22}}{Y_{21}}$$

$$B = -\frac{1}{Y_{21}}$$

$$C = \frac{-Y_{22} Y_{11} + Y_{12} Y_{21}}{Y_{21}}$$

$$D = \frac{-Y_{11}}{Y_{21}}$$

12. ABCD in terms of h parameters

$$V_1 = h_{11} I_1 + h_{12} V_2 \quad \text{--- (1)}$$

$$I_2 = h_{21} I_1 + h_{22} V_2 \quad \text{--- (2)}$$

From (2),

$$I_1 = \frac{1}{h_{21}} I_2 - \frac{h_{22}}{h_{21}} V_2 \quad \text{--- (3)}$$

sub (3) in (1) $\Rightarrow$

$$V_1 = \frac{h_{11}}{h_{21}} I_2 - \frac{h_{22} h_{11}}{h_{21}} V_2 + h_{12} V_2$$

$$= \frac{h_{11}}{h_{21}} I_2 + \left(\frac{-h_{22} h_{11} + h_{21} h_{12}}{h_{21}} \right) V_2 \quad \text{--- (4)}$$

From (3) + (4) $\Rightarrow$

$$A = \frac{-h_{22} h_{11} + h_{12} h_{21}}{h_{21}}$$

$$B = -\frac{h_{11}}{h_{21}}$$

$$C = -\frac{h_{22}}{h_{21}}$$

$$D = -\frac{1}{h_{21}}$$

Condition for reciprocity and symmetry in 2 port parameter representation

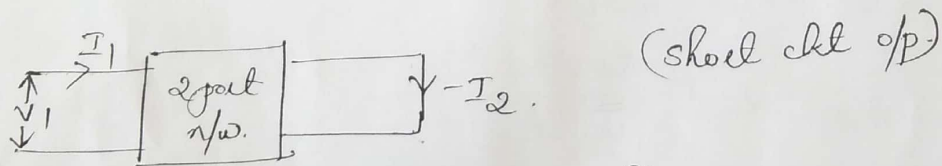
A network is said to be reciprocal if the ratio of the response variable to the excitation variable remains identical even if the positions of the response and excitation in the n/w are interchanged.

A two port network is said to be symmetrical if the i/p and o/p ports can be interchanged without altering the port voltages and currents. $\left[i.e., \frac{V_1}{I_1} = \frac{V_2}{I_2} \right]$

Z parameter representation

1. Condition for reciprocity

Case 1 :



$$V_1 = Z_{11}I_1 - Z_{12}I_2 \quad \text{--- (1)}$$

$$0 = Z_{21}I_1 - Z_{22}I_2 \quad \text{--- (2)}$$

$$\text{Ratio of response to excitation} = \frac{I_2}{V_1}$$

$$\text{From (2)} \Rightarrow I_1 = \frac{Z_{22}I_2}{Z_{21}}$$

sub this in (1) $\Rightarrow$

$$V_1 = \frac{Z_{11}Z_{22}}{Z_{21}}I_2 - Z_{12}I_2$$

$$\frac{I_2}{V_1} = \frac{Z_{21}}{Z_{11}Z_{22} - Z_{12}Z_{21}} \quad \text{--- (3)}$$

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Case 2

$$0 = -Z_{11}I_1 + Z_{12}I_2 \quad \text{--- (1)}$$

$$V_2 = -Z_{21}I_1 + Z_{22}I_2 \quad \text{--- (2)}$$

We need to find $\frac{I_1}{V_2}$

From (1),

$$I_2 = \frac{Z_{11}}{Z_{12}} I_1$$

sub this in (2) $\Rightarrow$

$$V_2 = -Z_{21}I_1 + \frac{Z_{22}Z_{11}}{Z_{12}} I_1$$

$$\frac{I_1}{V_2} = \frac{Z_{12}}{Z_{11}Z_{22} - Z_{21}Z_{12}} \quad \text{--- (4)}$$

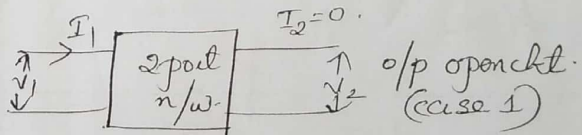
For the n/w to be reciprocal,

$$(3) = (4)$$

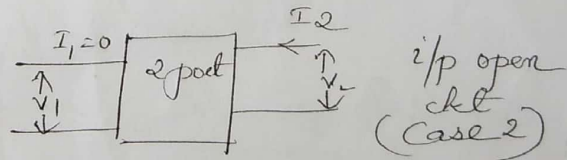
$$\therefore \underline{Z_{12} = Z_{21}} \quad \text{Condition for reciprocity.}$$

2. Condition for symmetry.

$$Z_{11} = \frac{V_1}{I_1}$$



$$Z_{22} = \frac{V_2}{I_2}$$

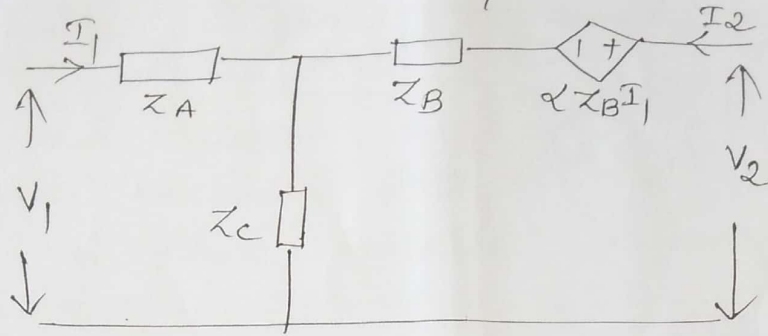


For the n/w to be symmetrical,

$$\frac{V_1}{I_1} = \frac{V_2}{I_2}$$

$$\therefore \underline{Z_{11} = Z_{22}}$$

Q.1) Obtain z parameters for the following ckt and show that the n/w is not reciprocal.



Sol:- Applying KCL in both the meshes,

$$V_1 = (Z_A + Z_C) I_1 + Z_C I_2 \quad \text{--- (1)}$$

$$V_2 = \alpha Z_B I_1 + (Z_B + Z_C) I_2 + Z_C I_1$$

$$= (\alpha Z_B + Z_C) I_1 + (Z_B + Z_C) I_2 \quad \text{--- (2)}$$

$$\therefore Z_{11} = Z_A + Z_C$$

$$Z_{12} = Z_C$$

$$Z_{21} = \alpha Z_B + Z_C$$

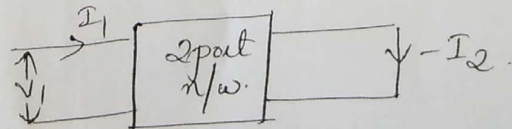
$$Z_{22} = \underline{\underline{Z_B + Z_C}}$$

Here, $Z_{12} \neq Z_{21} \quad \therefore$ The n/w is not reciprocal.

Condition for reciprocity & symmetry in y parameters.

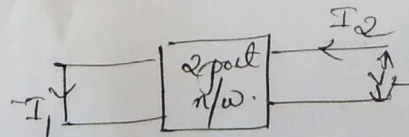
1. condition for reciprocity

$$-\frac{I_2}{V_1} = y_{21}$$



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$$-\frac{I_1}{V_2} = y_{12}$$



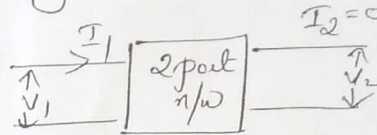
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For the n/w to be reciprocal,

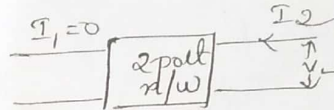
$$\underline{Y_{21} = Y_{12}}$$

2. Condition for symmetry

$$Y_{11} = \frac{I_1}{V_1}$$



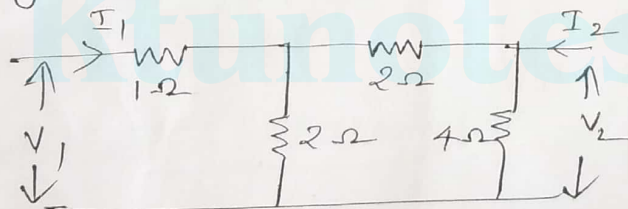
$$Y_{22} = \frac{I_2}{V_2}$$



For symmetrical ckt,

$$\underline{Y_{11} = Y_{22}}$$

1) Find Y parameters and check whether the ckt is symmetrical & reciprocal.



Ans: $Y_{11} = \frac{1}{4} \Omega$

$$Y_{21} = -\frac{1}{4} \Omega$$

$$Y_{12} = -\frac{1}{4} \Omega$$

$$Y_{22} = \frac{5}{8} \Omega$$

Here, $Y_{21} = Y_{12} \therefore$ N/w is reciprocal

$Y_{11} \neq Y_{22} \therefore$ N/w is not symmetrical.



h parameters

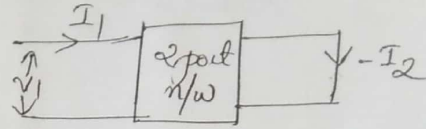
1) Condition for reciprocity

Case 1

$$V_1 = h_{11} I_1$$

$$-I_2 = h_{21} I_1$$

$$\Rightarrow \frac{I_2}{V_1} = -\frac{h_{21}}{h_{11}}$$



Case 2

$$0 = -h_{11} I_1 + h_{12} V_2$$

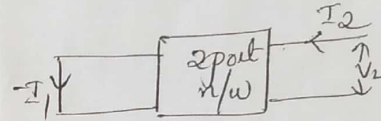
$$h_{11} I_1 = h_{12} V_2$$

$$\frac{I_1}{V_2} = \frac{h_{12}}{h_{11}}$$

For reciprocity,

$$\frac{I_2}{V_1} = \frac{I_1}{V_2}$$

$$\therefore \underline{h_{12} = -h_{21}}$$



2) Condition for symmetry

Case 1

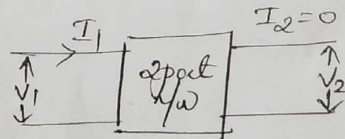
$$V_1 = h_{11} I_1 + h_{12} V_2 \quad \text{--- (1)}$$

$$0 = h_{21} I_1 + h_{22} V_2 \quad \text{--- (2)}$$

From (2),

$$V_2 = -\frac{h_{21}}{h_{22}} I_1$$

$$\text{sub in (1)} \Rightarrow V_1 = h_{11} I_1 - \frac{h_{12} h_{21}}{h_{22}} I_1$$



$$\frac{I_1}{V_1} = \frac{h_{22}}{\Delta h} \Rightarrow \frac{V_1}{I_1} = \frac{\Delta h}{h_{22}}$$

Case 2

$$V_1 = h_{12} V_2$$

$$I_2 = h_{22} V_2$$

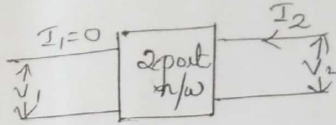
$$\frac{V_2}{I_2} = \frac{1}{h_{22}}$$

For n/w to be symmetrical,

$$\frac{V_1}{I_1} = \frac{V_2}{I_2}$$

$$\frac{\Delta h}{h_{22}} = \frac{1}{h_{22}}$$

$$\Delta h = 1$$



ABCD parameters

1. Condition for reciprocity

Case 1

$$V_1 = A V_2 - B I_2$$

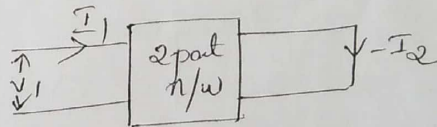
$$I_1 = C V_2 - D I_2$$

$$\Rightarrow V_1 = -B I_2$$

$$I_1 = -D I_2$$

$$\therefore \frac{-I_2}{V_1} = \frac{1}{B}$$

$$\frac{I_2}{V_1} = \underline{\underline{-\frac{1}{B}}}$$



Case 2

$$0 = AV_2 - BI_2 \quad \text{--- (1)}$$

$$I_1 = CV_2 - DI_2 \quad \text{--- (2)}$$

From (1),

$$I_2 = \frac{A}{B} V_2$$

$$\therefore I_1 = CV_2 - \frac{DA}{B} V_2$$

$$= \frac{BC - AD}{B} V_2$$

$$\frac{I_1}{V_2} = \frac{BC - AD}{B}$$

For the n/w to be reciprocal,

$$\frac{I_2}{V_1} = \frac{I_1}{V_2}$$

$$BC - AD = -1$$

$$\therefore \underline{\underline{AD - BC = 1}}$$

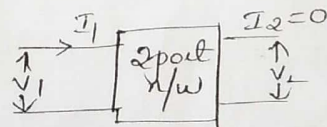
2. Condition for symmetry.

Case 1

$$V_1 = AV_2$$

$$I_1 = CV_2$$

$$\underline{\underline{\frac{V_1}{I_1} = \frac{A}{C}}}$$

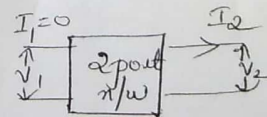


Case 2

$$V_1 = AV_2 - BI_2$$

$$0 = CV_2 - DI_2$$

$$\underline{\underline{\frac{V_2}{I_2} = \frac{D}{C}}}$$



For the n/w to be symmetrical,

$$\frac{V_1}{I_1} = \frac{V_2}{I_2}$$

$$\therefore \frac{A}{C} = \frac{D}{C}$$

$$\therefore \underline{A = D}$$

Parameter	Reciprocity	Symmetry
Z	$Z_{12} = Z_{21}$	$Z_{11} = Z_{22}$
Y	$Y_{12} = Y_{21}$	$Y_{11} = Y_{22}$
h	$h_{12} = -h_{21}$	$\Delta h = 1$
$ABCD$	$AD - BC = 1$	$A = D$

Different types of interconnections
of two port networks

Series Connection

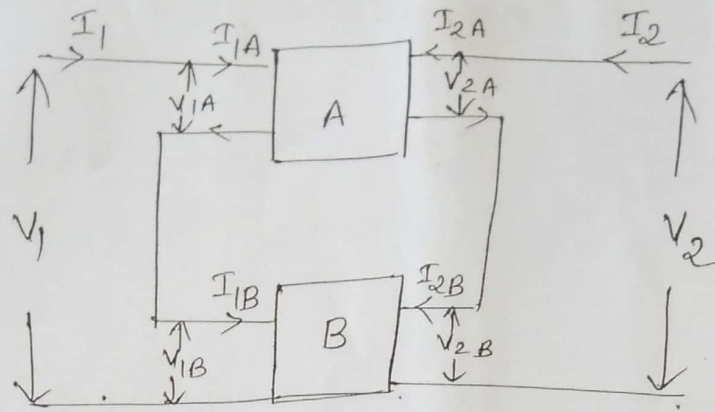
The open circuit impedance (Z) parameters is highly useful in characterising the series connected two port networks.

Here only two numbers of 2 port networks are shown to be connected in series, this result can be generalised for any number of two port n/w's in series.

Let n/w A and n/w B be the two port networks connected in series.



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From the fig it is clear that,

$$I_1 = I_{1A} = I_{1B}$$

$$I_2 = I_{2A} = I_{2B}$$

$$V_1 = V_{1A} + V_{1B}$$

$$V_2 = V_{2A} + V_{2B}$$

For n/w A,

$$V_{1A} = Z_{11A} I_{1A} + Z_{12A} I_{2A}$$

$$V_{2A} = Z_{21A} I_{1A} + Z_{22A} I_{2A}$$

For n/w B,

$$V_{1B} = Z_{11B} I_{1B} + Z_{12B} I_{2B}$$

$$V_{2B} = Z_{21B} I_{1B} + Z_{22B} I_{2B}$$

$$\text{Here, } V_1 = V_{1A} + V_{1B}$$

$$= Z_{11A} I_{1A} + Z_{12A} I_{2A} + Z_{11B} I_{1B} + Z_{12B} I_{2B}$$

$$= (Z_{11A} + Z_{11B}) I_1 + (Z_{12A} + Z_{12B}) I_2$$

$$V_2 = V_{2A} + V_{2B}$$

$$= (Z_{21B} + Z_{21A}) I_1 + (Z_{22A} + Z_{22B}) I_2$$

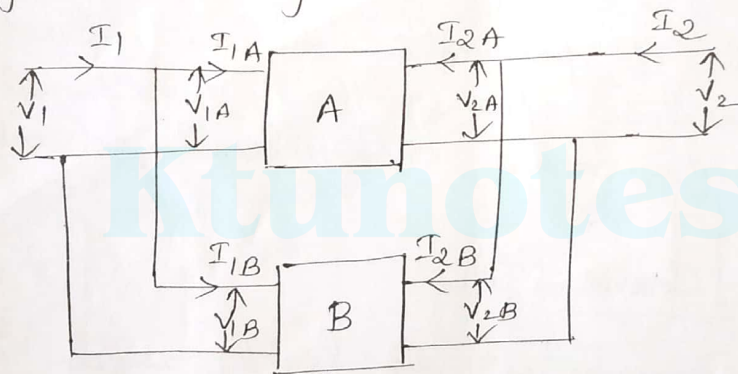
$$\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} Z_{11A} + Z_{11B} & Z_{12A} + Z_{12B} \\ Z_{21A} + Z_{21B} & Z_{22A} + Z_{22B} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix}$$

Note

The overall Z parameter matrix for series connected 2 port networks is simply the sum of Z matrices of each individual network.

Parallel Connection

Y parameter representation is used.



For n/w A,

$$I_{1A} = Y_{11A} V_{1A} + Y_{12A} V_{2A}$$

$$I_{2A} = Y_{21A} V_{1A} + Y_{22A} V_{2A}$$

For n/w B,

$$I_{1B} = Y_{11B} V_{1B} + Y_{12B} V_{2B}$$

$$I_{2B} = Y_{21B} V_{1B} + Y_{22B} V_{2B}$$

For parallel connection,

$$V_1 = V_{1A} = V_{1B}$$

$$V_2 = V_{2A} = V_{2B}$$

$$I_1 = I_{1A} + I_{1B}$$

$$I_2 = I_{2A} + I_{2B}$$

$$\begin{aligned} \therefore I_1 &= I_{1A} + I_{1B} \\ &= (Y_{11A} + Y_{11B}) V_1 + (Y_{12A} + Y_{12B}) V_2 \end{aligned}$$

$$\begin{aligned} I_2 &= I_{2A} + I_{2B} \\ &= (Y_{21A} + Y_{21B}) V_1 + (Y_{22A} + Y_{22B}) V_2 \end{aligned}$$

$$\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} Y_{11A} + Y_{11B} & Y_{12A} + Y_{12B} \\ Y_{21A} + Y_{21B} & Y_{22A} + Y_{22B} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix}$$

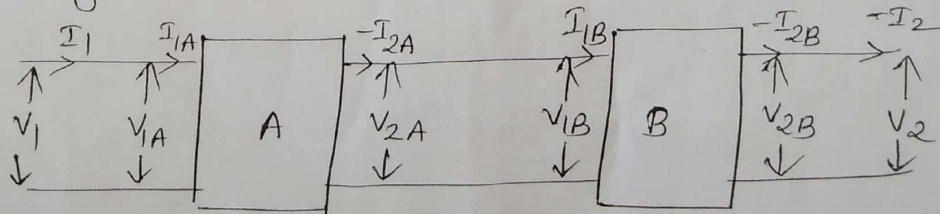
Note

The overall Y parameter matrix for parallel connected n/w is the sum of Y matrices of each individual network.

Cascade Connection

ABCD parameters are highly useful in characterising cascaded two port n/ws.

Let A and B be two n/ws connected in cascade. The result may be generalised for any no. of networks.



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For n/w A,

$$V_{1A} = A_A V_{2A} - B_A I_{2A}$$

$$I_{1A} = C_A V_{2A} - D_A I_{2A}$$

For n/w B,

$$V_{1B} = A_B V_{2B} - B_B I_{2B}$$

$$I_{1B} = C_B V_{2B} - D_B I_{2B}$$

For cascaded connection,

$$I_1 = I_{1A}$$

$$-I_{2A} = I_{1B}$$

$$I_2 = I_{2B}$$

$$V_1 = V_{1A}$$

$$V_{2A} = V_{1B}$$

$$V_2 = V_{2B}$$

$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} V_{1A} \\ I_{1A} \end{bmatrix} = \begin{bmatrix} A_A & B_A \\ C_A & D_A \end{bmatrix} \begin{bmatrix} V_{2A} \\ -I_{2A} \end{bmatrix}$$

$$= \begin{bmatrix} A_A & B_A \\ C_A & D_A \end{bmatrix} \begin{bmatrix} V_{1B} \\ I_{1B} \end{bmatrix}$$

$$= \begin{bmatrix} A_A & B_A \\ C_A & D_A \end{bmatrix} \begin{bmatrix} A_B & B_B \\ C_B & D_B \end{bmatrix} \begin{bmatrix} V_{2B} \\ -I_{2B} \end{bmatrix}$$

$$= \begin{bmatrix} A_A & B_A \\ C_A & D_A \end{bmatrix} \begin{bmatrix} A_B & B_B \\ C_B & D_B \end{bmatrix} \begin{bmatrix} V_1 \\ -I_2 \end{bmatrix}$$

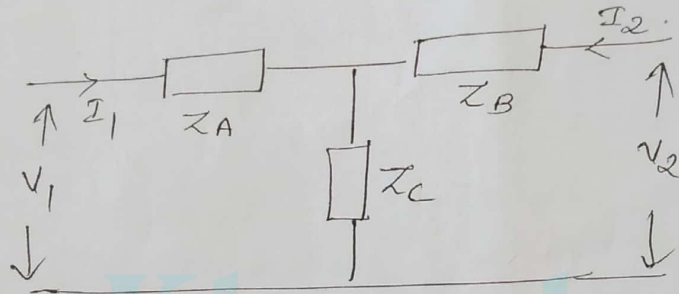


Note

The overall ABCD parameter n/w matrix for cascaded n/w is the matrix product of ABCD matrices of individual n/w.

Equivalent T and π section representation in parameter forms
T and π n/ws are frequently used to represent the equivalence of tm lines, filters etc

T representation



To find Z parameters,
Applying KVL in both meshes,

$$V_1 = (Z_A + Z_C) I_1 + Z_C I_2$$

$$V_2 = (Z_B + Z_C) I_2 + Z_C I_1$$

$$\therefore Z_{11} = Z_A + Z_C$$

$$Z_{12} = Z_{21} = Z_C$$

$$Z_{22} = Z_B + Z_C$$

$$\therefore Z_A = Z_{11} - Z_{12}$$

$$Z_B = Z_{22} - Z_{12}$$

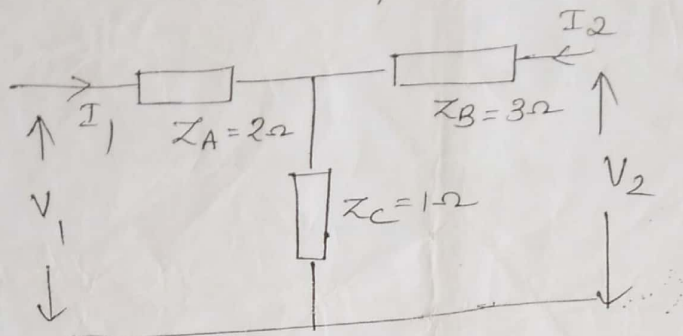
$$Z_C = Z_{12}$$

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Q. Draw the eqt T n/w representation.
 Given $Z_{11} = 3\Omega$, $Z_{12} = Z_{21} = 1\Omega$, $Z_{22} = 4\Omega$

Ans:-

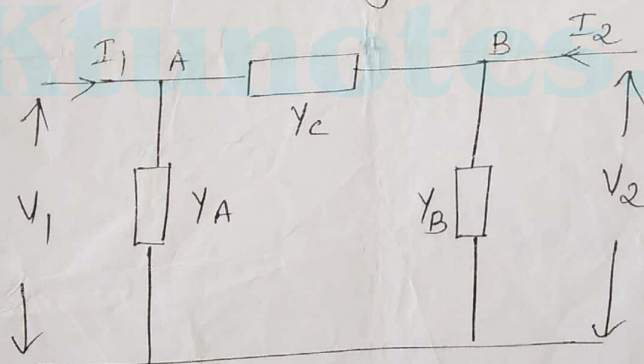


$$Z_A = Z_{11} - Z_{12} = 3 - 1 = 2\Omega$$

$$Z_B = Z_{22} - Z_{12} = 4 - 1 = 3\Omega$$

$$Z_C = Z_{12} = 1\Omega$$

Equivalent π representation



To find Y parameters, apply KCL at node A & B.

$$I_1 = V_1 \cdot Y_A + (V_1 - V_2) \cdot Y_C = (Y_A + Y_C)V_1 - Y_C V_2$$

$$I_2 = V_2 \cdot Y_B + (V_2 - V_1) \cdot Y_C = (Y_B + Y_C)V_2 - Y_C V_1$$

$$\therefore Y_{11} = Y_A + Y_C$$

$$Y_{12} = Y_{21} = -Y_C$$

$$Y_{22} = Y_B + Y_C$$



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$$\therefore Y_A = Y_{11} + Y_{12}$$

$$Y_B = Y_{22} + Y_{12}$$

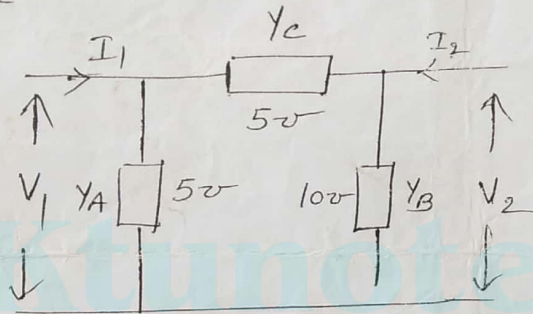
$$Y_C = -Y_{12}$$

Q. Draw the eqt. π n/w representation where $Y_{11} = 10 \text{ S}$, $Y_{12} = Y_{21} = -5 \text{ S}$, $Y_{22} = 15 \text{ S}$

Ans:- $Y_A = Y_{11} + Y_{12} = 5 \text{ S}$

$$Y_B = Y_{22} + Y_{12} = 10 \text{ S}$$

$$Y_C = -Y_{12} = 5 \text{ S}$$

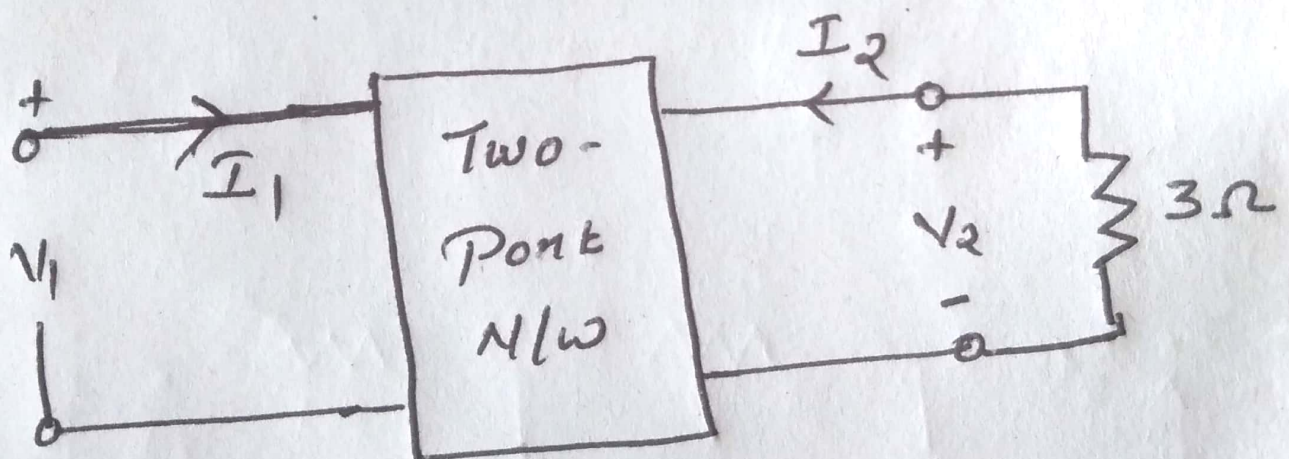


Q) The following eqn gives the voltage and current at the i/p port of a two port n/w shown in figure

$$V_1 = 5I_1 + 2I_2$$

$$V_2 = 2I_1 + I_2$$

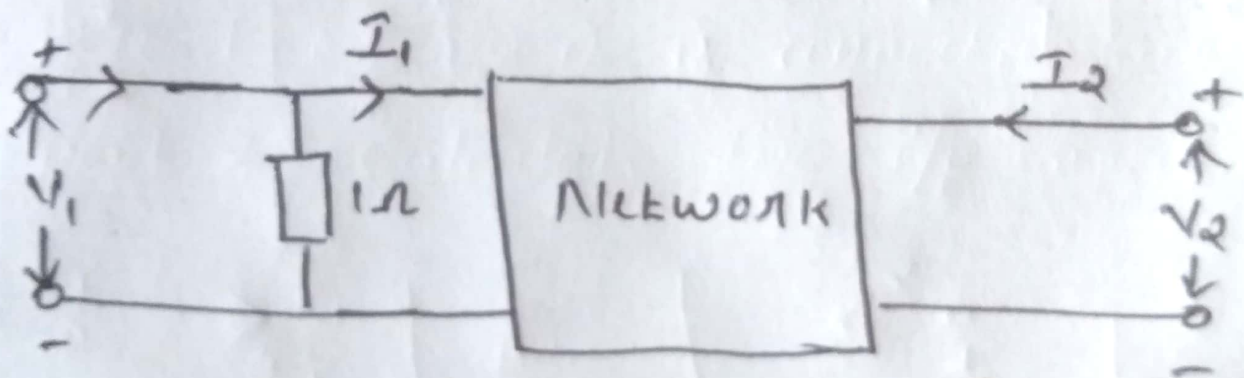
A load of 3Ω is connected across port 2. Calculate the i/p impedance.



Q) The y -parameters for a two-port n/w shown in fig are given as

$$y_{11} = 4\text{ S}, y_{22} = 5\text{ S}, y_{12} = y_{21} = 4\text{ S}$$

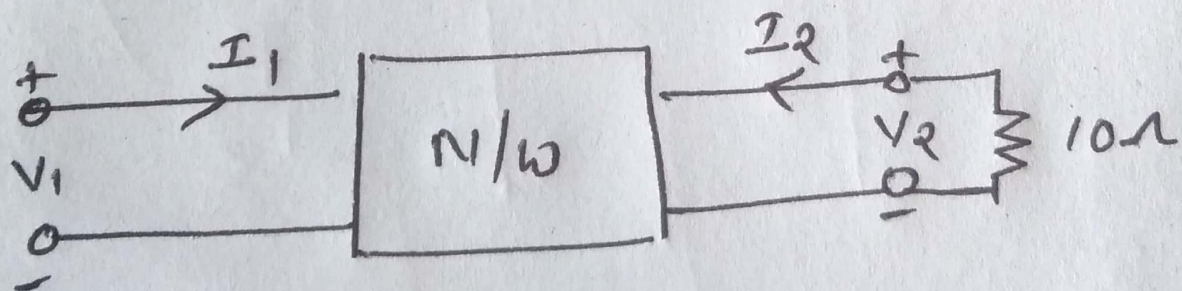
If a resistor of 1Ω is connected across port-1 of the network then find the output impedance.



Q) The ABCD parameters of a 2-port n/w is shown in figure and

$$A = 2.5, B = 4 \Omega, C = 1 \text{ V}, D = 2$$

What must be the i/p voltage V_1 applied for the o/p voltage V_2 to be 10V across the load of 10Ω connected at port 2?



Q)

Find y and z -parameters of the n/w shown in figure?

